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Ochanomizu University

Complement Control without a DP Controller

Asako Matsuda

Synopsis

The generative literature has often focused on the obligatory nature of complement control. This means that the referent of one of the matrix DPs is fully identical with that of the null subject of the complement clause. However, there are various complement control cases, such as split control, partial control, and implicit dative control, which do not entertain full identity between the syntactically represented matrix DP and the null complement subject. The present paper argues that a complement control relation is a V-CP relation, and that we can dispense with a DP controller. Based on Japanese data, I propose that the controlled complement has its own subject-like referential property associated with its own force. Thus, the null subject does not have to depend on a matrix DP for its reference. I illustrate that the relevant referential properties are best captured by the *speaker/addressee* features. My proposal naturally accounts for split control, partial control, and implicit control as well as the canonical subject and object control. It also explains the *de se* reading characteristic of complement control.*

Keywords: *obligatory control, split control, partial control, de se reading, force*

1. Introduction

The generative literature has treated complement control in various ways. Some have viewed it as a DP-DP relation between a matrix DP and a null infinitival subject (Chomsky (1981), Manzini (1983)). Some have analyzed it as an indirect DP-DP relation, in which Agr at C mediates the controller-PRO relation (Borer (1989), Landau (2000)). Another type of analysis has proposed a movement theory that treats control as a DP-trace relation (Hornstein (1999, 2003)). Yet another type has argued that it is a DP-VP relation between a matrix DP and a complement VP (Williams (1980), Chierchia (1984), Dowty (1985)). However, despite the divergence of these views, the matrix DP controller has almost always been considered to be indispensable in a control relation. The common assumption was and still is that a controlled complement clause lacks its own subject with an independent referential property, so that it

must have an outside DP controller.

Views like the above are rooted in the observation that complement control canonically involves obligatory control,¹ in which, by definition, the referent of one of the matrix DPs is fully identical with that of the null subject of the complement clause (Williams (1980)); the obligatory control criteria require a unique, local (i.e., located in the immediately higher clause), and c-commanding controller. However, there are various complement control cases which do not meet such criteria. Consider the following. (1) and (2) are adapted from Landau (2000), and (3) from Koster (1984).²

- (1) John_i proposed to Mary_j [PRO_{i+j} to meet each other at 6]. (split control)
- (2) Mary recommended to John_i [PRO_{i+} to convene without her]. (partial control)
- (3) My teacher suggested (to me_i) [PRO_i to take another topic]. (implicit dative control)

In a split control case such as (1), there is no single matrix DP that fully controls PRO. In a partial control case like (2), the matrix DP *John* constitutes only a subset of the reference of PRO. (3) exhibits implicit dative control, in which the controller is absent from the matrix clause. Faced with these facts, one gets tempted to question if an outside DP controller is a prerequisite for complement control.

The purpose of this paper is to argue that a complement control relation is a V-CP relation, and that we can dispense with a DP controller. The DP controller does not have to be syntactically present in the sentence, just as the referent of *you* in a sentence like *you are beautiful* does not have to be syntactically represented. This is a possibility because, as I will argue, the controlled complement has its own subject-like referential property to begin with. This may sound senseless at first since there are no signs whatsoever of a referential subject within English controlled complements. However, there are some signs in other languages that suggest the existence of a referential property within the controlled complement. Thus, control without a DP controller is a plausible option that deserves serious consideration.³

In providing evidence that complement control can do away with a DP controller, I will mainly focus on English and Japanese in this article. However, my proposal is not intended to explain certain phenomena in these particular languages. The major goal of the present study is to propose a crosslinguistically generalizable account for complement control.

This paper proceeds as follows. Section 2 presents an argument that there are signs of a referential subject within a complement clause, and the subject is associated with the clausal force. Such an argument is supported by Japanese data. Section 3 discusses the nature of the referential complement subject, followed by section 4, which deals with how clausal forces of the controlled complements are semantically selected by the control predicates. After these basic ideas are introduced, section 5 provides the overall picture of my proposal. Sections 6 and 7 present empirical and theoretical consequences of my proposal, respectively. The last section concludes this paper.

2. Signs of a Referential Subject within a Complement Clause

This section provides some evidence that controlled complement clauses have an independent referential

subject-like element within its own clause. The following Japanese data show overt force morphology, and remarkably, the relevant forces are inherently associated with certain person features such as *the speaker* and *the addressee*. It turns out that, what appears to be a subjectless complement in fact has a referential element, which feeds into PRO to serve as a subject.

2.1. Independent Force in Complement Clauses

First, observe the following Japanese data with a distinct postverbal particle in each embedded clause.⁴

- (4) Taro_i-wa [PRO_i daigaku-e iki-tai-to] negat-ta. (subject control)
Taro-Top [university-to go-Opt-C_{to}] hope-Past
'Taro hoped to go to university.'
- (5) Taro_i-wa [PRO_i daigaku-e ik-oo-to] kuwadate-ta. (subject control)
Taro-Top [university-to go-Int-C_{to}] attempt-Past
'Taro attempted to go to university.'
- (6) Hanako_i-wa Taro_j-ni [PRO_j daigaku-e ik-e-to] meireisi-ta. (object control)
Hanako-Top Taro-Dat [university-to go-Imp-C_{to}] order-Past
'Hanako ordered Taro to go to university.'
- (7) Hanako_i-wa Taro_j-ni [PRO_i daigaku-e ik-u-to] yakusokusi-ta. (subj. acr. obj.)
Hanako-Top Taro-Dat [university-to go-Prm-C_{to}] promise-Past
'Hanako promised Taro to go to university.'
- (8) Hanako_i-wa Taro_j-ni [PRO_{i+j} daigaku-e ik-oo-to] teiansi-ta. (split control)
Hanako-Top Taro-Dat [university-to go-Exh-C_{to}] propose-Past
'Hanako proposed to Taro to go to university (together).'

The complement clause in (4) appears with the optative (Opt) particle *-tai*, exhibiting subject control. In (5), the intensive (Int) particle *-yoo* occurs, and it also shows subject control. (6) has the imperative (Imp) particle *-e-ro* in the complement, bringing about object control, while (7) appears with the *-ru* promissive (Prm) particle, giving rise to subject control across object. Lastly, (8) occurs with the exhortative (Exh) particle *-yoo*, and split control is observed.⁵

What we find in the data is that each sentence, itself a declarative, has a complement clause inside it with an independent force, as schematized in (4)' to (8)'. Furthermore, there seems to be a correlation between the complement force and the control pattern.⁶

- (4)' [Decl. *hope* [Opt.]] => subject control
(5)' [Decl. *attempt* [Int.]] => subject control
(6)' [Decl. *order* [Imp.]] => object control
(7)' [Decl. *promise* [Prm.]] => subject control across object
(8)' [Decl. *propose* [Exh.]] => split control

Before we move on to see what this correlation amounts to, some words are in order for each associated force particle.

Optative

In Japanese, the optative particle *-tai* appears freely in both root and embedded environments. It is used to express one's desire, expectation, and willingness to do something. However, there is an important restriction in the usage of this particle. As illustrated by the data below adapted from Sugawara (2005), it is used only with a first person subject in a statement, although it could be used with a second person subject in a question.

- (9) Watasi-wa susi-o tabe-tai.
I-Top sushi-Acc eat-Opt
'I want to eat sushi.'
- (10) Anata-wa nani-o tabe-tai?
you-Top what-Acc eat-Opt
'What do you want to eat?'
- (11) *Anata/*Kare-wa susi-o tabe-tai.
you/he-Top sushi-Acc eat-Opt
Intended: 'You/He want(s) to eat sushi.'

Only the speaker can make an affirmative statement about his/her own feeling, or ask the addressee about the addressee's own feeling. A second or third person subject cannot cooccur with the *-tai* optative in an affirmative statement for this reason.

Intentive

Fujii (2010) appropriately describes the intentive particle *-yoo* to be associated with "essentially internal monologue-like utterances" (p.215). Typically, the particle is used in an utterance to oneself to stay resolute on something. For example, a school child may mutter (12) to himself/herself coming home from school to avoid the temptation to put it off until the next day.

- (12) Kyoo shukudai si-yoo. (monologue)
Today homework do-Int
'I'll do my homework today.'

Since the intentive particle is used in a monologue or to show overhearers that one is resolute on doing something, it only appears with a subject that refers to the speaker, hence the ungrammaticality of (13).

- (13) *Anata/*Kare-wa kyoo shukudai si-yoo.
you/he-Top today homework do-Int
Intended: 'You'll/He'll do your/his homework today.'

Imperative

Japanese imperatives, which appear with the imperative particle *-e/-ro*, are used in a very similar way as those of English. For instance, a father may utter (14) to his son.

- (14) Kyoo shukudai si-ro.
Today homework do-Imp
'Do your homework today.'

As you might expect, the subject of an imperative is always the addressee. It cannot appear with a subject that refers to the speaker or some other third person. Note the ungrammaticality of (15).

- (15) *Watasi/*Kare-wa kyoo shukudai si-ro.
I/he-Top Today homework do-Imp
Intended: 'I/he must do my/his homework today.'

Promissive

Although promissives are not crosslinguistically as prevalent as imperatives according to Portner (2004), Korean is attested to have promissives, and I assume that Japanese also has promissives; the particle *-ru* is used to mark this force. Consider the following:

- (16) Watasi-ga kopi-o tor-u-yo.⁷
I-Nom copy-Acc take-Prm/Nonpast-Prt
'I'll make copies.'
- (17) Kare-ga kopi-o tor-u-yo.
he-Nom copy-Acc take-?Prm/Nonpast-Prt
'He'll make copies.'

We can interpret (16) in at least two ways: one is a plain prediction that the speaker will make copies (*-ru* as a nonpast tense marker); the other is a promissive in which the speaker makes a promise that he/she will make copies (*-ru* as a promissive). The latter makes the speaker be obligated to do something. In contrast, for (17), the only available reading is a prediction; the speaker cannot make a promise concerning some other person's action under normal circumstances. It may be interpreted as a promissive in a special context, in which the speaker is responsible for someone else's action: when *kare* 'he' is the speaker's son, student, etc. It is uncertain at this point why a promissive reading with a third person subject is allowed in such a context. I leave this to be explored in a future study. The present study focuses

on the fact that a promissive reading is not available with a third person subject without a special context. Note that the acceptability even declines with a second person subject in (18) for both promissive and predictive readings:

- (18) Anata-ga kopi-o tor-u-yo.
 you-Nom copy-Acc take-*Prm/?Nonpast-Prt
 ‘You’ll make copies.’

This illustrates that the *-ru* particle most naturally gives rise to a promissive reading when the speaker is the subject of the sentence.

Exhortative

Lastly we turn to the exhortative, *-yoo*. This particle is phonologically identical with the intensive particle. However, I follow Fujii (2006) in distinguishing them as two different particles. When used as an exhortative, it functions in nearly the same way as ‘let’s’ in English; a child utters (19) to ask his/her friends to work on their assignment together.

- (19) Kyoo shukudai si-yoo. (talking to friends)
 Today homework do-Exh
 ‘Let’s do our homework today (together).’

The first person plural inclusive of the second person (i.e., *we* meaning ‘I and you,’ not *we* meaning ‘I and some other third person’) is the only appropriate subject this particle can cooccur with. Using it with the other subjects brings about ungrammaticality.

- (20) *Anatatati/*Karera-wa kyoo shukudai si-yoo.
 you.pl./they-Top today homework do-Exh
 Intended: ‘Why don’t you/they do your/their homework today (together).’

I have shown that the embedded clause in a complement control construction in Japanese appears with an overt force particle, in particular, with an optative, intensive, promissive, imperative, or exhortative particle. The distribution of the force particles revealed that each force is associated with a particular subject. This suggests that the controlled complement clause may have a certain subject-like referential element associated with its own force.

Zanuttini, Pak, and Portner (2012) propose that the clausal force with person restrictions is hosted in some head within the CP domain although they do not mention precisely which head. In this paper, I tentatively presuppose a non-split CP, and assume that the single C head hosts the force and the associated referential elements.

2.2. Do English Complement Clauses Have Their Own Force?

English controlled clauses show no signs of a referential subject. Does it mean, then, that they do not have their own force that is associated with a certain subject?

Consider the following English data, which parallel the Japanese data just observed above.

- (21) Taro_i hoped [PRO_i to go to university].
- (22) Taro_i attempted [PRO_i to go to university].
- (23) Hanako ordered Taro_i [PRO_i to go to university].
- (24) Hanako_i promised Taro [PRO_i to go to university].
- (25) Hanako_i proposed to Taro_j [PRO_{i+j} to go to university].

As expected, we see no force distinctions in them; all embedded clauses appear with a *to*-infinitive. Nonetheless, I do not take this fact to demonstrate no existence of the force; what we see is a syncretism in English force marking. Thus, the configurations (4)' to (8)' in the previous subsection also hold for these English cases, and in the counterparts of many other languages. The morphology remains null for these forces in some languages, but they do exist syntactically within the complement clauses. As I will argue below, complement control facts fall out very naturally under such an assumption.

2.3. Force and the Subject in the Previous Literature

I owe it to the notion of the To-Do List proposed by Portner (2004) that I have come to associate the clausal force with its subject. Portner, by introducing the notion of the To-Do List, successfully characterizes some sentential forces. According to him, To-Do List is a set of properties. Certain forces have the discourse function of adding a property to the To-Do List of the speaker and/or the addressee. To be more precise, the discourse function of imperatives is to add the property denoted by the predicate to the To-Do List of the addressee. For example, an imperative such as *shut the door!* adds the property of *shutting the door* onto the To-Do List of the addressee. Similarly, the force of promissives would be to add some property to the speaker's To-Do List, and the force of exhortatives, to the To-Do List of both the speaker and the addressee.

Following Portner, Zanuttini et al. (2012) thoroughly investigate imperatives, promissives, and exhortatives and their restrictions on the subject. They maintain that these three clausal forces, *Jussives* as they call them, are syntactically and semantically very similar. They focus on Korean data, in which all three forces are attested, and successfully reveal the association of the force and the subject: imperatives restrict their subject to be the addressee(s); promissives restrict it to be the speaker(s) exclusive of the addressee(s); and exhortatives the speaker(s) inclusive of the addressee(s). The Japanese data we saw above snugly pattern with their observation on Korean *Jussives*.

I assume that the Japanese intensitive is another type of *Jussive*; it adds a property to the To-Do List of the speaker. However, with regard to the optative, I take a different view; it is not a type of *Jussive*, and does not involve Portner's To-Do List. An important contrast I observe between the optative and the other relevant forces is that the optative can denote both intentional actions (e.g., *run the marathon*) and

nonintentional situations (e.g., *resemble your father*), while the others are only felicitous with intentional actions. The optative does not have to denote some actions for someone *to do* in a way that the other *Jussives* do. It simply states one's desire, expectation or willingness. This said, I still assume that the optative imposes restrictions on the subject as we saw above.

The following table summarizes the restrictions on the subject by force. The feature representations I will use in this paper are shown in the last column.

Table 1. Subject Restrictions by Force

Force		Subject Restrictions	
optatives		The speaker(s) exclusive of the addressee(s)	(+SP, -AD)
<i>Jussives</i>	intentives	The speaker(s) exclusive of the addressee(s)	(+SP, -AD)
	imperatives	The addressee(s) exclusive of the speaker(s)	(-SP, +AD)
	promissives	The speaker(s) exclusive of the addressee(s)	(+SP, -AD)
	exhortatives	The speaker(s) inclusive of the addressee(s)	(+SP, +AD)

3. The Nature of the Person Feature

This section deals with the question why the person feature specified in terms of the speaker/addressee can be inherently referential. First, imagine that John utters the following sentence to Mary.

(26) Taro persuaded Hanako [_{CP} C_{Imp} (-SP, +AD) to study abroad].

We identify the complement clause to be an imperative. This leads to assuming its C head to be hosting the person feature (-SP, +AD), so that we consider the subject of the complement to be the addressee(s). In the context I provide, John is the speaker and Mary is the addressee. This may lead one to think my system returns Mary to be the complement subject, contrary to the fact. Indeed, it does not return Mary to be the subject, because the speaker/addressee notion I employ here is not identical to the standard notions of the first- and the second-person pronouns.

So-called pronouns such as English *I*, *we*, *you*, *he*, *she*, *it*, and *they* are often argued to be decomposable (Postal (1970), Déchaine and Wiltschko (2002), Harley and Ritter (2002), van Koppen (2012)). Pronouns seem to be at least made up of the determiner component and the noun component. For example, the pronoun *I* consists of the abstract definite determiner *I* and the noun content *one*. Thus, at some abstract level, the pronoun *I* is represented as *Ione* and *you* as *youone* (Postal (1970)). In this sense, I consider the speaker and the addressee features to be equivalent to the definite determiner components of *I* and *you*. They are the first- and the second-person counterparts of the third person definite determiner *the* in English.

Such a definite determiner-like nature of the speaker/addressee feature allows, and in fact requires, a *shifted* interpretation⁸ within an embedded environment, contrasted with the full DP pronouns *I* and *you*, whose referents are fixed to the speaker and the addressee of the speech time context. Put another

way, the specific referent of the speaker/addressee feature is determined relative to the event context of the immediately higher clause. In (26), the matrix clause denotes the event in which Taro is persuading Hanako; in such a context, the speaker is Taro and the addressee Hanako (note that they are not in italics). The addressee feature hosted in the C head refers to Hanako this way. If the full DP pronoun *you* appears in the complement clause as in (27), it refers to the addressee of the speech context; if John utters (27) to Mary, *you* refers to Mary.

(27) Taro convinced Hanako that *you* should study abroad.

Thus far, the contrast between the speaker/addressee features and the full DP first- and second-person pronouns has been emphasized, but they are similar in that they both have an independent referential property. From (26), it seems as though the addressee feature is referentially dependent on the matrix DP *Hanako*, but the question is: is it syntactically dependent? Observe the following root imperative.

(28) [_{CP} C_{Imp} (-SP, +AD) Shut up]!

I contend that the same addressee feature is present in root imperatives like (28). In (28), there is no higher clause. Yet, under normal circumstances, if we are to be present at the time and place where this imperative is uttered, we know who the subject is, by observing the discourse context, i.e., by observing who the speaker is looking at, who stopped talking after the utterance and so on. I maintain that we are doing the exact same thing when we figure out the referent of the embedded addressee feature in a control construction like (26). It is not the matrix DP *Hanako* which is the referent of the addressee feature. We just come to interpret the referent of the addressee to be Hanako, by imagining the event described by the matrix clause. We are capable of imagining such an event because we know the semantics of the predicate *persuade*, in addition to our world knowledge associated with the act of persuasion.

Thus, the speaker/addressee feature is not syntactically dependent on the matrix DP. It has an independent and inherent referential capacity. In terms of the referential capacity, it is very similar to the regular pronouns such as *I* and *you*, the sole difference being the availability of a shifted reading. We tend to assume that the interpretation of a sentence like *I love you* comes so handy with no complicated assumptions. However, to know what this really means, one has to know who uttered this sentence to whom; one has to be able to imagine how the event of someone uttering this sentence took place. Only after that, one finds out the specific referents of *I* and *you* in this sentence. We are doing precisely the same thing with the embedded speaker/addressee feature in complement control constructions. For these reasons, I strongly hold that the speaker/addressee feature has an independent referential property.

4. S-selection and Argument Structure

The discussion so far in a way argued that the complement clause is much more independent from its

matrix clause than it has traditionally been assumed. In fact, such independence of the complement clause constitutes the core argument of my proposal in this paper. However, this does not imply that the matrix and the complement clauses are completely independent from each other. They are related in some limited ways.

I argue that they are related in two ways. First, they are contextually related in the way I discussed in the previous section. Second, I observe a certain correlation between the matrix predicate and the complement force: the predicates such as *hope* and *prefer* appear with an optative; *attempt* and *intend* cooccur with an intensitive; *promise* and *vow* with a promissive; *order* and *persuade* with an imperative; and *propose* typically appears with an exhortative. A similar pattern is also detected in Japanese. This section focuses on this second relation.

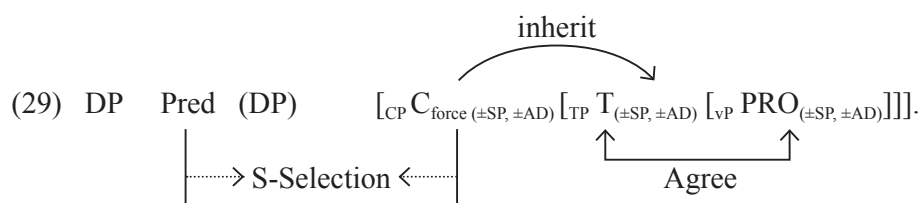
I contend that the notion of semantic selection (s-selection) introduced in Grimshaw (1979) accounts for such a correlation. Grimshaw posits that the predicates like *ask* and *wonder* s-select interrogatives while those like *amazing* and *be surprised at* s-select exclamatives. There are also some predicates that select both semantic types (e.g. *know*, *find out*) or neither (e.g. *think*). The present study holds that just like the predicate *wonder* s-selects an interrogative, the control predicates such as *order* s-select an imperative, *promise*, a promissive, and so on. They do not have to be a biunique relation; some predicates select multiple forces. For example, the predicates such as *propose* and *offer* appear to s-select imperative, promissive and exhortative clauses, giving rise to variable control.

The next question is: how do control predicates s-select the complement force? I assume that such information is encoded in the argument structure of the predicate in the lexicon. The argument structure has often been perceived to be responsible for assigning particular thematic roles to the DP arguments. For instance, in the argument structure of the predicate *promise*, the *source* role is assigned to its external argument and the *goal* role to its internal argument. However, there is another internal argument, which has always been assumed to be present in the argument structure, but its specific *role* has not attracted much attention (except the declarative-interrogative distinction). That is the internal propositional argument, i.e. the complement clause. It may be plausible to assume that the internal propositional argument is assigned a role in terms of its force such as optative, intensitive, and promissive. Just like the thematic roles such as *source*, *goal*, *agent*, *patient*, and *theme* are assigned to the DP arguments, specific forces may be assigned to the propositional arguments.

The fact that this proposal is dependent on the lexicon may make it look much like some other thematic approaches to control. However, my proposal should not be confounded with them. While other thematic approaches require both the argument structure of the predicate and some kind of control module to associate the structure with the control pattern, mine only requires the argument structure. For instance, Farkas (1988) introduces the *Resp* relation, and Jackendoff and Culicover (2003) and Culicover and Jackendoff (2006) bring in the notion of *conceptual structure*-binding relation to understand the control patterns. This is because their focus is on the thematic roles of the argument DPs; this requires them to bring in another set of semantic rules to associate the specific DP role to the agent (i.e. the null subject) of the action denoted by the complement clause. This has led to a very complicated and convoluted theory of control, and how such semantic rules map onto the syntactic structure has never been clearly illustrated.

5. Proposal

Based on these arguments, I propose the derivation of complement control to proceed as schematized in (29):



This proposal crucially meets the spirit of minimalism. First, the Phase Impenetrability Condition (PIC, Chomsky (2001)) is fully respected. Only the C head has to be visible during the derivation of the immediately higher cycle where the matrix predicate is introduced. The approaches with a DP controller have often suffered from a theoretical shortcoming in which the controller DP was criticized to be “too far” from the complement subject (i.e. PRO).

I believe there are no previous theories of complement control that achieve such simplicity without serious empirical cost. In the following two sections, I will discuss empirical and theoretical consequences of this proposal.

My proposal naturally accounts for split control, implicit control, and partial control as well as the canonical subject and object control. Additionally, it explains the *de se* reading characteristic associated with control.

Consider (21) to (25), repeated here as (30) to (34) with new annotations.

- All these control patterns are natural consequences of my proposal. Let's consider (34), a split control case. The complement clause is an exhortative, meeting the selectional requirement of the matrix predicate *propose*. The C head of the complement clause hosts the exhortative force with its designated person feature, (+SP, +AD). Through C to T inheritance and T-PRO Agree, PRO ends up with the (+SP, +AD) feature, which is independently referential. The referent of PRO is identified by consulting with the matrix event context. It turns out that *Hanako* and *Taro* (precisely, they should not be indicated in italics) are the referents of PRO.

Implicit dative control poses no problem to my proposal. Consider (35), repeating (3) with new annotations.

(35) My teacher suggested (to me)_[CP C_{Imp} (-SP, +AD)] PRO_(-SP, +AD) to take another topic].

In (35), the complement clause is an imperative, which meets the selectional requirement of the predicate *suggest*. The derivation assigns PRO with the (-SP, +AD) feature, indicating that the addressee of the matrix context is the referent. It could be anybody assumed to be the addressee in the matrix context. It can be the speaker of this whole utterance as indicated in the parenthesis (*me*) if he/she is understood to be the receiver of the teacher's suggestion from the context. It can also be Taro if Taro is understood to be the addressee of the matrix event, i.e. the receiver of the suggestion, from the previous conversation.

The most important point is that my proposal does not require a syntactically represented DP controller. Indeed, the present study explains the observation made by Rizzi (1986): even a structurally unrepresented element (i.e. a no-*pro* element) can control. This falls out naturally if we view a control relation as a V-CP relation and assume that a DP controller does not have to be syntactically projected. In this system, all we need is the person feature represented in C of the complement clause, which has its own referential property. It refers to the matrix event context, not to the matrix DP, to identify its referent. It turns out that implicit control is not only unproblematic for my proposal, but actually serves as strong evidence for it.

6.3. Partial Control

Consider (36), repeating (2). The addressee of the matrix context, the referent of *John*, constitutes only a subset of the reference set of PRO.

(36) Mary recommended to John_i [PRO_{i+} to convene without her].

Partial control is explainable under my proposal by appealing to the notion of *associative semantics* inherent in the speaker and the addressee person indexicals. As is often noted, the first- and the second-person plural forms do not always refer to multiple speakers or addressees. Instead, they refer to a group of people, in which the speaker or the addressee is included, hence, *associative semantics*, according to Wechsler (2010). Wechsler views the associative nature of the speaker/addressee indexicals to be an absolute universal, which is supported by crosslinguistic evidence. Although many different variations of personal pronoun systems have been attested in the world's languages, some logical alternatives are not attested. Consider the following table (Table 2) adapted from Wechsler (2010:335). I added the last column to show the correspondence with the speaker/addressee features I employ in my proposal.

Table 2. The Seven Logically Possible Meta-persons, and the Four Attested Pronoun Types

Possible		Attested	(±SP, ±AD)
1+2 1+2+3	speaker(s) and addressee(s); no others speaker(s), addressee(s), and other(s)	A. 'inclusive'	(+SP, +AD)
1 1+3	speaker(s) only speaker(s) and other(s); addressees excluded	B. 'exclusive'	(+SP, -AD)
2 2+3	addressee(s) only addressee(s) and other(s); speakers excluded	C. 'second person'	(-SP, +AD)
3	other(s) only	D. 'third person'	(-SP, -AD)

In the table, 1 stands for the speaker, 2 the addressee, and 3 any non-participant in the speech act. There are seven logically possible combinations, or 'meta-persons,' indicated in the first column, but languages seem to make only four-way distinctions (A to D) at maximum as shown in the third column. This implies that apart from D, 'third person,' there is no pronominal category attested that prohibits the inclusion of 'other' non-participants or requires the inclusion of them. The attested categories A, B, and C all can, but need not, include 'other' non-participants. This reveals that associative semantics is inherent in the speaker/addressee person indexicals. Importantly, the speaker/addressee features I adopt in this study are such indexicals with an inherent associative nature.

Now the partial control fact observed in (36) follows naturally. I assume that it has the following configuration (37):

(37) Mary recommended to John [_{CP} C_{Imp} (-SP, +AD) PRO_(-SP, +AD) to convene without her].

Here, the complement is an imperative clause. Thus, PRO ends up with the feature (-SP, +AD). This person feature on PRO has the same property as the category C pronoun in Table 2. The referent can be a single addressee, multiple addressees, or addressee(s) plus others, but it prohibits the inclusion of the speaker.

The person feature associated with the force only narrows down the range of interpretation of PRO to be what falls under a certain category of Table 2. It does not go any further to identify precisely who is to be interpreted as PRO. Such precise identification is dependent on other factors, i.e., the discourse context or the complement predicate. In the case of (37), the complement predicate *convene* requires a collective or plural subject. The addressee in the matrix context is understood to be *John* (more precisely, the referent of *John*), but *John* alone does not constitute a collective or plural subject. This brings about an addressee-plus-others reading.

In fact, if (35) in the previous subsection is uttered by a student who is preparing a group presentation, PRO can have a plus-others reading. Such prevalence of partial control is extensively studied in Landau (2000), although he explains this fact differently, as I will discuss later.

The dependence on the context and the complement predicate in precise identification of PRO may make my argument appear inconclusive. However, I argue that the interpretation of any pronoun is dependent on the context and/or the predicate. Observe the following:

(38) We will convene at nine.

In (38), *we* can refer to a single speaker plus multiple addressees, a single speaker plus multiple addressees and others, or a single speaker plus others. There are even more ways to interpret *we* in (38). However, *we* in (38) perhaps does not refer to a single speaker and a single addressee, because the predicate *meet* rather than *convene* is normally preferred to express the situation in which two individuals get together. The interpretation of pronouns is always dependent on the context and/or the predicate. Thus, the ambiguity of PRO in my proposal is not unique to PRO. It is the inherent nature of all pronouns.

6.4. Control Shift

I argue that there are two routes for control shift: one is rooted in the argument structure of the control predicate; the other resorts to the pragmatic process *coercion*. Control shift observed in (39) taken from Rooryck (2000) falls under the first group.

(39) Kim_i proposed to Sandy_j [PRO_{i/j/i+j} to do the dishes].

A sentence like the above is ambiguous in three ways: PRO can be interpreted to refer to Kim or Sandy or both Kim and Sandy. It exhibits control shift or variable control. I maintain that this is possible because the predicate *propose* semantically selects three different force types, promissive, imperative, and exhortative. In other words, there are three different predicates taking the same form *propose*, each corresponding to a distinct argument structure. In one structure, the predicate s-selects the promissive force, whereas in another one, it s-selects the imperative, and in yet another one, it s-selects the exhortative.

The other type of control shift involves coercion. The example in (40), cited in Landau (2000), typically shows this type of control shift.

(40) Susie_i persuaded the teacher [PRO_i to be allowed to leave early].

Here, the predicate *persuade*, which normally gives rise to object control manifests subject control. I assume such a shift to be a consequence of a selectional mismatch between the matrix predicate and the complement force. Intentives, imperatives, promissives, and exhortatives (but not optatives) fall under what Zanuttini et al. (2012) call *Jussives* as discussed above, and crucially, they should only denote *intentional* actions. In (40), however, the complement clause denotes a nonintentional situation. Thus (40) brings about a selectional mismatch, triggering coercion. According to Jackendoff and Culicover (2003), coercion is “a conventionalized omission of semantic material in syntactic expression” (p.542). In many cases, the semantic material ‘allow/enable X to’ is omitted from the shifted sentences. Under coercion, (40) can be paraphrased as follows:

(41) Susie persuaded the teacher to enable her to be allowed to leave early.

Importantly, my proposal makes the trigger of coercion to be syntactically visible. It sees the coercion involved in control shift to be a consequence of a selectional mismatch between two syntactic sisters, the matrix predicate and the complement force, which is represented in the C head of the complement. In fact, such an account for control shift is almost in line with some previous literature such as Sag and Pollard (1991) and Jackendoff and Culicover (2003). They also focus on the argument structure of the matrix predicate and the complement's semantic content to explain the shift. However, their arguments do not illustrate how the semantic relations are made syntactically visible. The present study certainly builds on these previous works, but departs from them by taking a V-CP approach and by adopting the notions of complement force and s-selection, which serve as a bridge between semantics and syntax

6.5. *De Se* Reading

Consider the following well-known scenario discussed in Higginbotham (1992) based on a similar story in Castañeda (1967). A war hero suffering from amnesia (referred to as “the unfortunate man”) reads a story of his own wartime heroic deeds, which he unfortunately does not remember. He thinks that this hero should get the Medal of Honor.

(42) The unfortunate man_i expects that he_i should get the Medal of Honor.

(43) The unfortunate man_i expects [PRO_i to get the Medal of Honor].

(42) correctly describes the situation whereas (43) does not. This is because PRO in (43) only allows a *de se* reading contrasted with the overt pronoun in (42) allowing both *de se* and *de re* readings.

The speaker/addressee features directly account for such an asymmetry between the pronoun *he* and PRO. According to Wechsler (2010), the speaker/addressee person indexicals⁹ inherently have the force of bringing about a self-ascription or reference *de se*. I assume that the speaker/addressee features adopted here are such person indexicals, and thus, they are the source of the obligatory *de se* reading associated with the relevant complement control constructions.

The reason that a structure like (42) can bring about both *de se* and *de re* readings is because there is no particular self-ascriber in the structure. It remains mute on whether *the unfortunate man* knows the referent of *he* is in fact himself. As a result, both *de se* and *de re* readings remain available.

7. Theoretical Consequences

This section discusses what theoretical consequences my approach may have with respect to some prominent minimalist approaches to control, namely, the movement theory of control (MTC) and the Agree theory of control (ATC).

The MTC (Hornstein (1999, 2003), Boeckx and Hornstein (2003, 2004), Hornstein and Polinsky (2010))¹⁰ perceives a control relation to be a movement chain, in which the head matrix DP is the controller and the null complement subject is its trace. The ATC (Landau (2000, 2004, 2006, 2008))

pursues an indirect C-mediated DP-DP approach to complement control, which dates back to Borer (1989).

Because both of them assume a complement control relation to involve a DP controller, neither overcomes the shortcomings inherently associated with such a view. As presented in the introduction, complement control cases abound that do not entertain full identity between one of the matrix DP arguments and PRO. Neither the MTC nor the ATC straightforwardly accounts for such cases.

The MTC, although theoretically minimal, only accounts for simple subject control (as in (30) and (31)) and object control (as in (32)). The MTC proponents treat the other complement control patterns as *peripheral* or nonexistent. Hornstein (2003) explicitly rules out split control, and admits that partial control does not follow from the MTC. As to the *promise*-type subject control, Hornstein (2003) and Hornstein and Polinsky (2010) offer an account that assumes a null preposition before the object; this way, the object does not count as an intervener blocking the movement. However, such an account is not very convincing, since there are predicates that explicitly appear with a preposition (e.g. *propose to*, *signal (to)*) and give rise to object control. With regard to control shift, they argue that “the postverbal DP previously treated as a thematic object comes to be interpreted as a thematic oblique indirect object” (Hornstein and Polinsky (2010:22)). Nevertheless, the MTC has not explained what motivates the shift in the thematic structure to bring about the shift in control.

Additionally, Hornstein and Pietrosky (2010), taking the MTC approach, argue that their copy-and-merge approach to movement works as a vehicle for *de se* readings in obligatory control. They posit that the controller and the controllee being identical copies brings about the obligatory *de se* reading. However, I wonder how a proper name, such as *John*, suddenly bears a self-ascribing capacity when being copied. If the expression *John* does not self-ascribe, then its identical copy should not either. I believe my proposal is more convincing in this respect, in that it adopts the notion of person indexicals that are inherently self-ascribers.

Landau’s ATC does not fare better in most cases. The ATC is proposed specifically for partial control. Landau considers partial control to be more prevalent than it has long been assumed, and thus, believes that a theory for it is in order. Indeed, my proposal owes greatly to his descriptive analyses of partial control, but we take different approaches to explain it. Landau argues that there are two routes of control: one is for exhaustive control (EC) predicates such as *manage*, *fail*, *begin*, and *continue*; the other is for partial control (PC) predicates such as *intend*, *prefer*, *hope*, and *promise*.¹¹ He assumes two Agree operations for both EC and PC. In EC, one Agree operation takes place between the probe T (if subject control) or the probe v (if object control) of the matrix clause and the goal, the DP controller; and the other Agree relation holds between the same probe and the goal, PRO. On the other hand, in PC, one Agree operation holds between the probe T/v of the matrix clause with the goal, the DP controller; and the other Agree occurs between the same probe with the goal, which is not PRO but T-agr of the complement clause raised to C. T-agr also enters an Agree relation with PRO. Through these multiple Agree operations, the ϕ -features are properly shared between the DP controller and PRO. What distinguishes PC from EC in this system is how semantic plurality is interpreted through these Agree operations; in PC, T-agr at C mediates the Agree relation, and this makes the semantic plurality on PRO to be invisible to the probe, Landau argues. Thus, a partial control relation, in which PRO is semantically plural but the controller is

non-plural, becomes a possibility.

Nonetheless, the ATC only allows partial control; it does not explain it, nor does it account for which matrix argument partially controls PRO. Besides, it cannot account for split control, control shift, or implicit control. In fact, Landau seems to assume that syntax and non-syntactic factors (i.e. semantics and pragmatics) work independently in control constructions. He maintains that syntax is involved in controller determination in “delimiting a potential domain — namely, the domain of the matrix clause,” but the choice of controller within that domain is “sensitive to complex semantic/pragmatic factors” (Landau 2000:197). His ATC only takes care of the syntactic part, which just accommodates but does not explain certain control facts. The rest is left to independent semantic/pragmatic mechanisms. His arguments do not make clear how semantic and pragmatic factors are mapped onto his syntactic system.

Returning to the MTC, it seems to have a certain empirical advantage in that it can account for backward control and adjunct control. To see if the basic view proposed in the present article can systematically explain these phenomena is definitely on my future research agenda. However, the MTC’s empirical coverage with respect to the types of complement control studied here is extremely narrow; thus, on the whole, I do not consider its advantage to be indisputable. The central aim of the MTC is “grammatical downsizing” (Hornstein (1999:69)), but it seems to me that it pays measurable empirical price for such downsizing. The present proposal at least achieves comparable empirical coverage, and thus, deserves as serious consideration as the MTC.

To sum up, these minimalist approaches to control, the MTC and the ATC, do not have an obvious edge over my proposal. Empirically, more complement control patterns become explainable under the present approach than these previous ones. Theoretically, my proposal is much simpler than the ATC, and at least as minimal as the MTC. It should serve as an alternative minimalist approach to complement control, worthy of earnest investigation.

8. Conclusion

This paper introduced a new approach to complement control, which dispenses with a DP controller. The approach is fundamentally distinct from previous generative approaches. No other previous theories have seriously dealt with a similar approach. Although the present study is only at a nascent stage, and the proposal is not yet fully developed, it seems that we are moving towards the right direction. A whole variety of complement control facts just fall out so naturally under the proposed assumptions.

My proposal can be viewed as a synthesis of syntactic and semantic approaches to complement control, but it fundamentally departs from the traditional approaches by denying the existence of a DP controller. This has allowed the proposal to dispense with the control module once conceived to constitute a vital part of the theory of control. As a consequence, my proposal has become theoretically minimal, and at the same time, its empirical coverage has broadened.

I believe that pursuing the same approach in future studies will contribute to unearthing some essential aspects of language associated with the phenomena of control.

Endnotes

- * I am very grateful to Professor Tohru Noguchi and the two anonymous reviewers for their constructive and insightful comments that helped to improve the manuscript considerably. I also wish to thank Professor Edward Schaefer for carefully proofreading the manuscript and revising the English. Needless to say, all the remaining errors are my own.
- 1 Control into interrogative complements, commonly viewed as non-obligatory control, is outside the scope of this paper. Whether and how the present proposal remains consistent with interrogative control will be studied in future research.
- 2 These patterns of control have been issues in the study of control for quite some time: Jackendoff (1972) observed split control, Wilkinson (1971) partial control, and Kimball (1971) implicit control.
- 3 The basic ideas leading to a V-CP approach to control are developed in Matsuda (2014), which views complement control as clause type selection and complementizer agreement.
- 4 Both direct and indirect speech readings are available for these embedded clauses (see Fujii (2006:110-114)). Here, I focus on the indirect speech reading.
- 5 The Japanese optative particle, *-tai*, follows the verb in a continuative form, ‘renyookei.’ The particle *-yoo* (both intensive and exhortative) is realized as *-yoo* when it follows vowel-final stems and as *-oo* when it follows consonant-final stems, and similarly, the imperative particle appears as *-e* after consonant-final stems and as *-ro* after vowel-final stems (Fujii (2006:100)). Likewise, the particle *-ru* is realized as *-u* when it follows consonant-final stems and as *-ru* when it follows vowel-final stems.
- 6 Fujii (2006, 2010) observes a similar correlation.
- 7 This sentence sounds most natural with the discourse particle *-yo*; the sentence sounds very abrupt without this particle, although it is still acceptable. According to Endo (2007), this particle “implies the speaker’s emphasis in giving a piece of information to the addressee” (p.180).
- 8 I borrow this terminology from Delfitto and Fiorin (2011).
- 9 Indexicals depend on the utterance context for their reference. Wechsler states that first- and second-person pronouns “are ESSENTIAL in the sense that they cannot be semantically reduced to nonindexical descriptions” (Wechsler 2010:341).
- 10 Some other movement approaches to control had already been proposed before Hornstein (1999) such as Martin (1996) and O’Neil (1997).
- 11 I speculate that most of Landau’s EC predicates are restructuring predicates, thoroughly explored in Wurmbrand (2001).

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